CS & IT ENGINEERING

2026

Computer Network

Transport Layer

Lecture No. - 07

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Recap of Previous Lecture

Topic TCP Data transfer

Topics to be Covered

Topic

TCP Connection Close

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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#Q. Consider a TCP connection in a state where there are no outstanding ACKs. The sender sends two segments back to back. The sequence numbers of the first and second segments are 230 and 290 respectively. The first segment was lost, but the second segment was received correctly by the receiver. Let X be the amount of data carried in the first segment (in bytes), and Y be the ACK number sent by the receiver. The values of X and Y (in that order) are :

[GATE-2007]

(A) 60 and 290

(B) 230 and 291

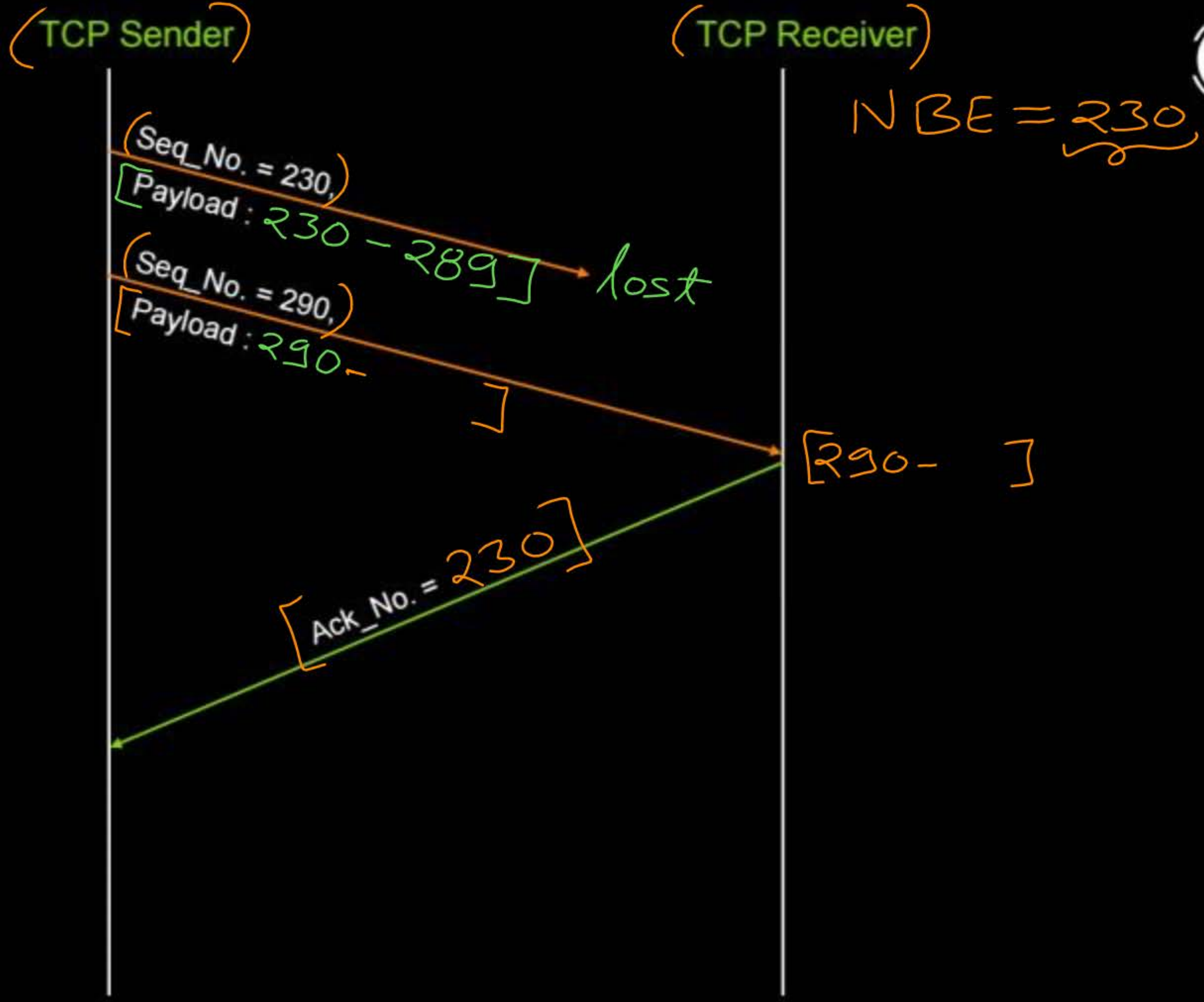
(C) 60 and 231

(D) 60 and 230

X = Amount of Data carried in first seg.

X = (290 − 230) = 60 byte

$\boxed{\text{Ans: D}}$



TCP Sender
TCP Receiver
NBE = 230
Seq_No. = 230,
Payload : 230 - 289
lost
Seq_No. = 290,
Payload : 290-
290-
Ack_No. = 230

[MSQ] [GATE-2023][2 Mark]

#Q. Suppose you are asked to design a new reliable byte-stream transport protocol like TCP. This protocol, named myTCP, runs over a 100 Mbps network with Round Trip Time of 150 milliseconds and the maximum segment lifetime of 2 minutes. Which of the following is/are valid lengths of the Sequence Number field in the myTCP header?

A. ~~30 bits~~

B. 32 bits

C. 34 bits

D. 36 bits

$\boxed{\text{Ans: B, C \& D}}$

Solution :-

Maximum number of bytes can be transmitted in one MSL = MSL * Bandwidth

$= 2\ min * 100\ Mbps = 2 \times 60\ sec * 10^8\ bits/sec$

$= 12 * 10^9\ bits \qquad = \left(\frac{12 * 10^9}{8}\right)$ bytes

Minimum number of bits required for sequence number field

= ⌈log₂(Max no. of bytes can be transmitted in one MSL)⌉

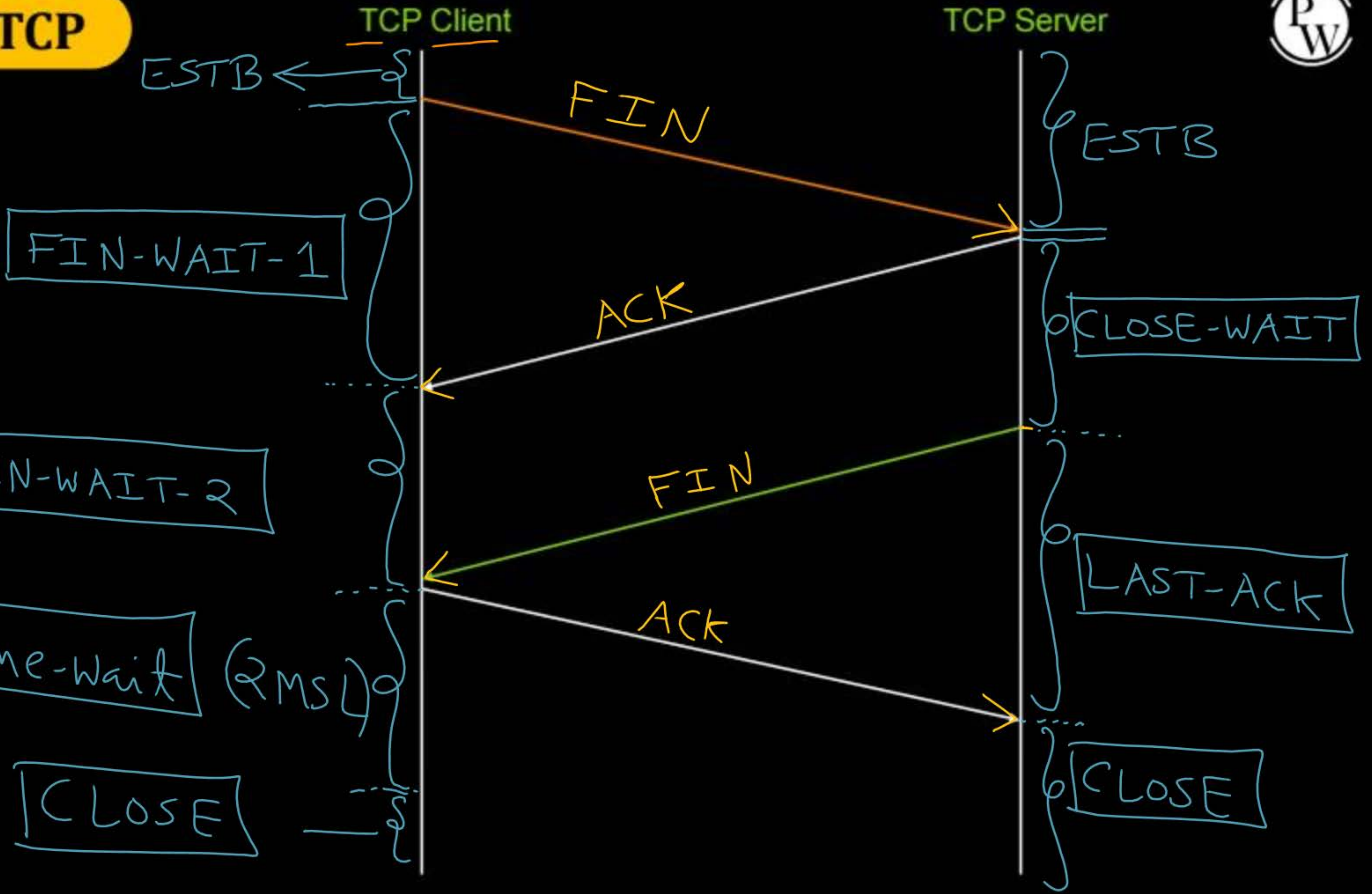
$= \left\lceil \log_2\left(\frac{12 * 10^9}{8}\right)\right\rceil$ bits $= \lceil 30.48 \rceil$ bits $= \boxed{31\ bits}$

Topic : TCP Connection Close

→ Connection close between TCP client and TCP server

→ 4-way handshake process

→ TCP client $\boxed{\text{or}}$ TCP server any one can initiate the connection close request


TCP Client
TCP Server
ESTB
FIN
ESTB
FIN-WAIT-1
ACK
CLOSE-WAIT
FIN-WAIT-2
FIN
LAST-ACK
ACK
Time-wait (2MSL)
CLOSE
CLOSE

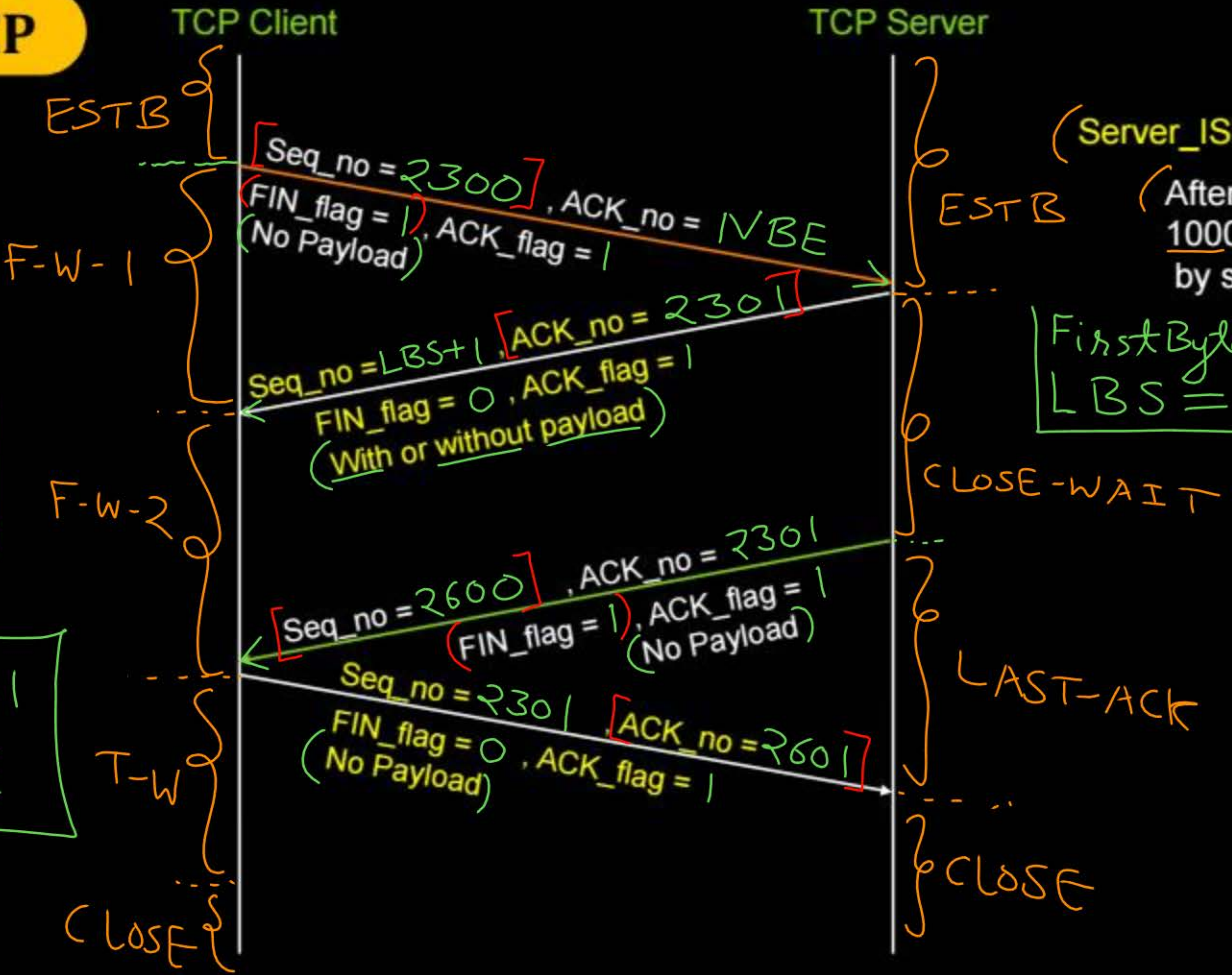
Topic : TCP

(Client_ISN = 299)

(After sending 2000 bytes by client.)

FirstByte = 300
LastByteSent = 2299

Seq. No. = LBS + 1
Ack No. = NBE



(Server_ISN = 1599)

(After sending 1000 bytes by server.)

FirstByte = 1600
LBS = 2599

Topic : TCP

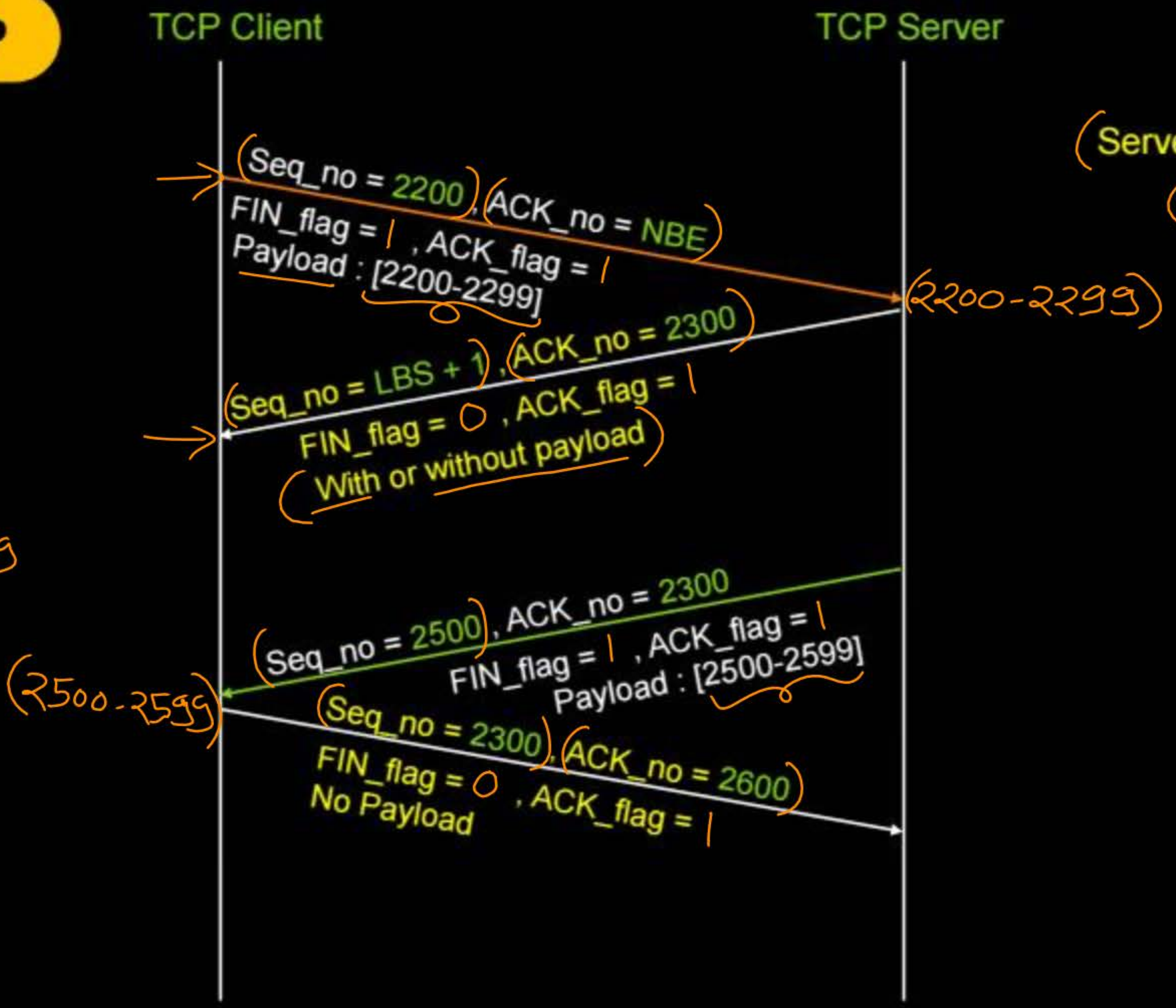
Client_ISN = 299

MSS = 100 bytes

Client sending 2000 bytes.

1900 bytes transmitted

LBS = ~~2199~~ 2299

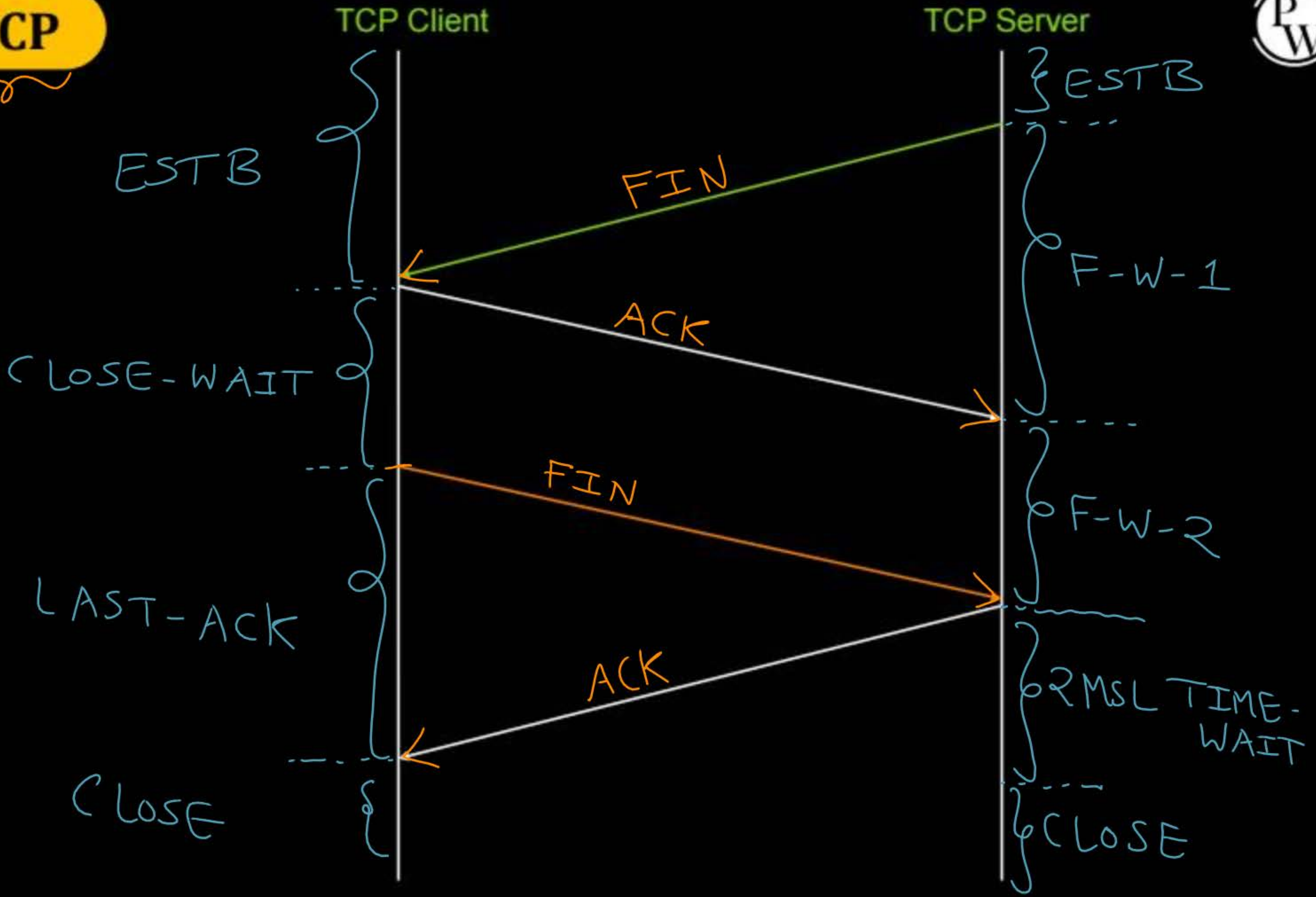


Server_ISN = 1599

Server sending 1000 bytes.

900 byte transmitted

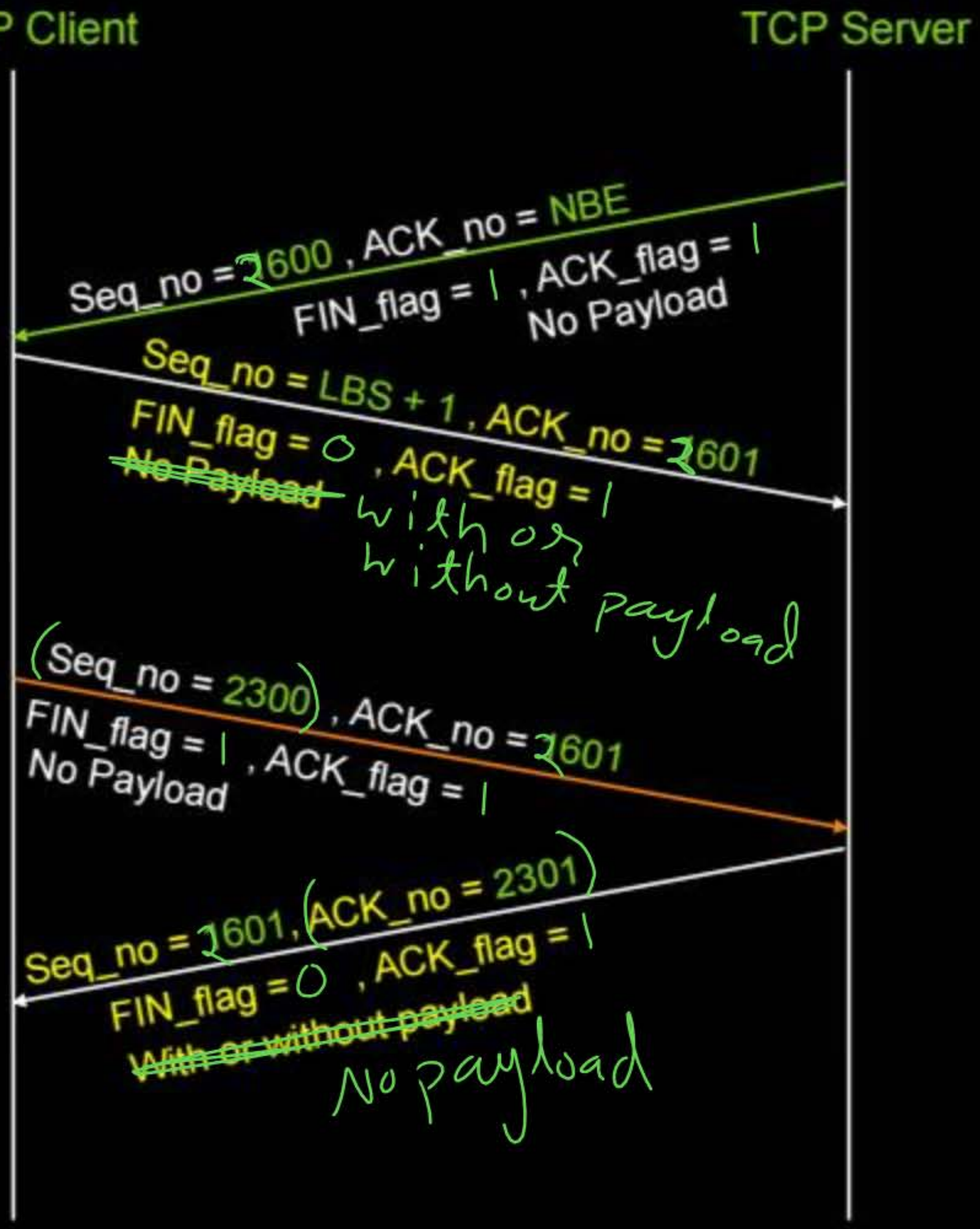
LBS = 2499


TCP Client
TCP Server
ESTB
ESTB
FIN
F-W-1
ACK
CLOSE-WAIT
FIN
F-W-2
LAST-ACK
ACK
2MSL TIME-WAIT
CLOSE
CLOSE

Topic : TCP

Client_ISN = 299

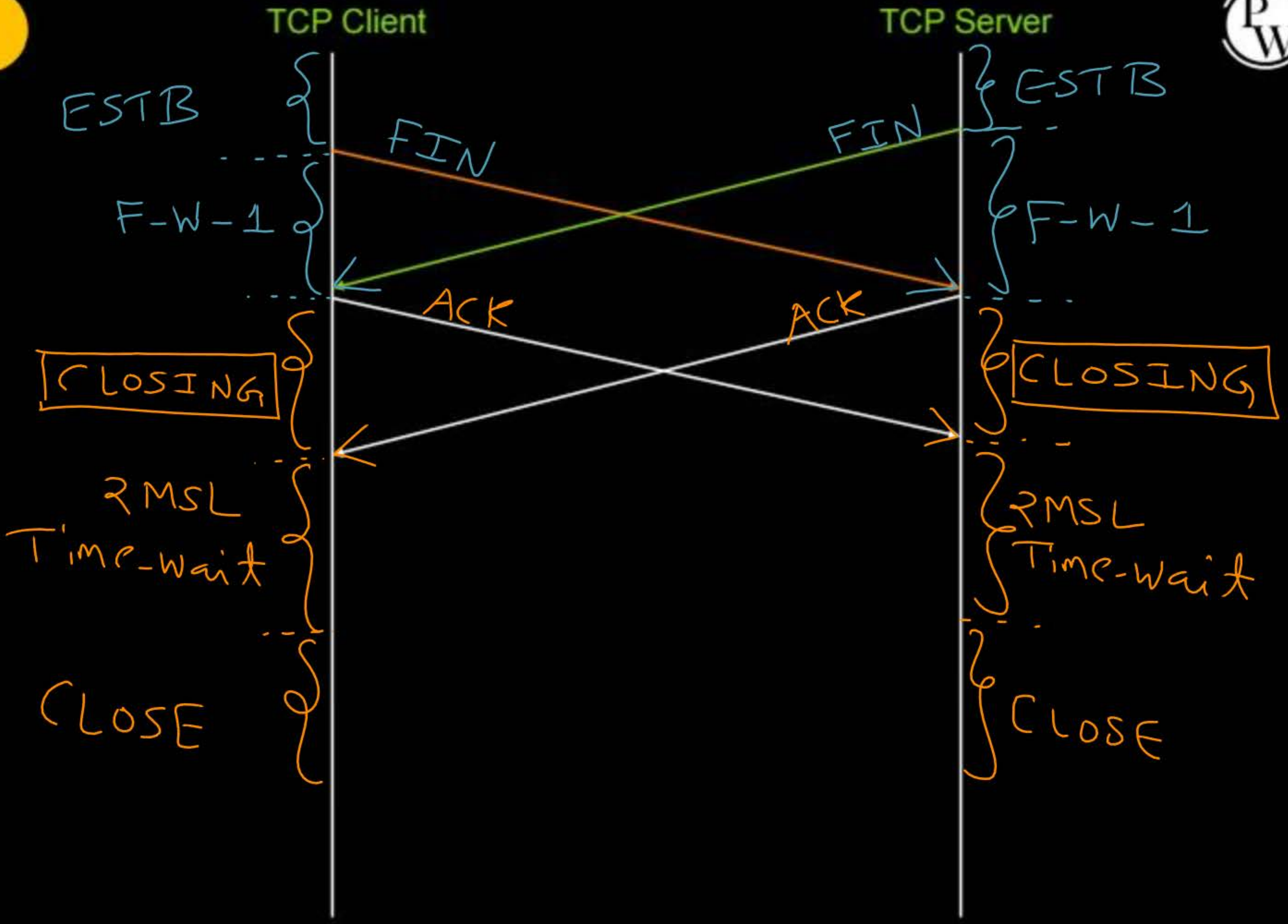
After sending 2000 bytes by client.



Server_ISN = 1599

After sending 1000 bytes by server.

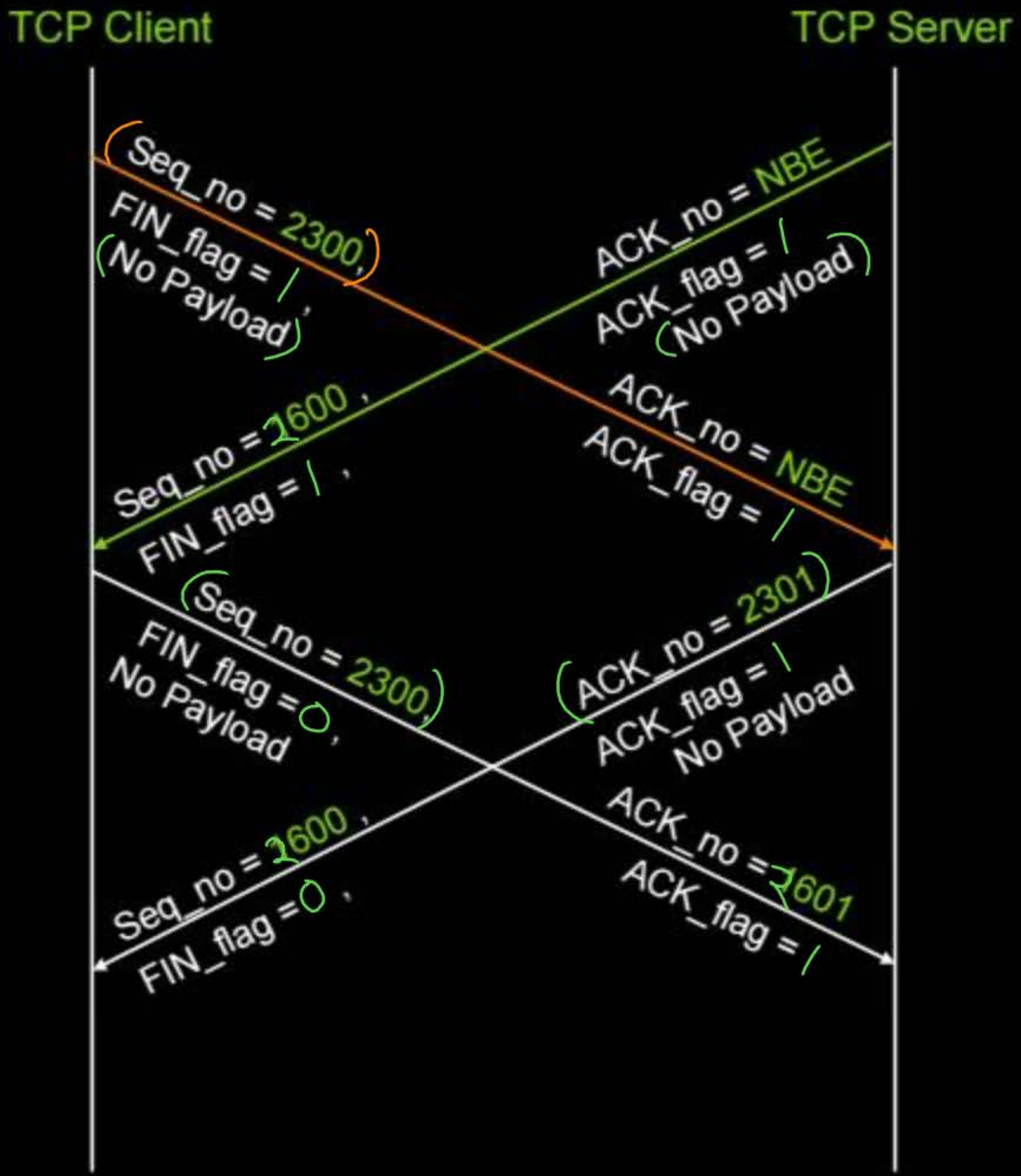
Topic : TCP


TCP Client
TCP Server
ESTB
ESTB
FIN
FIN
F-W-1
F-W-1
ACK
ACK
CLOSING
CLOSING
2MSL
Time-wait
2MSL
Time-wait
CLOSE
CLOSE

Topic : TCP

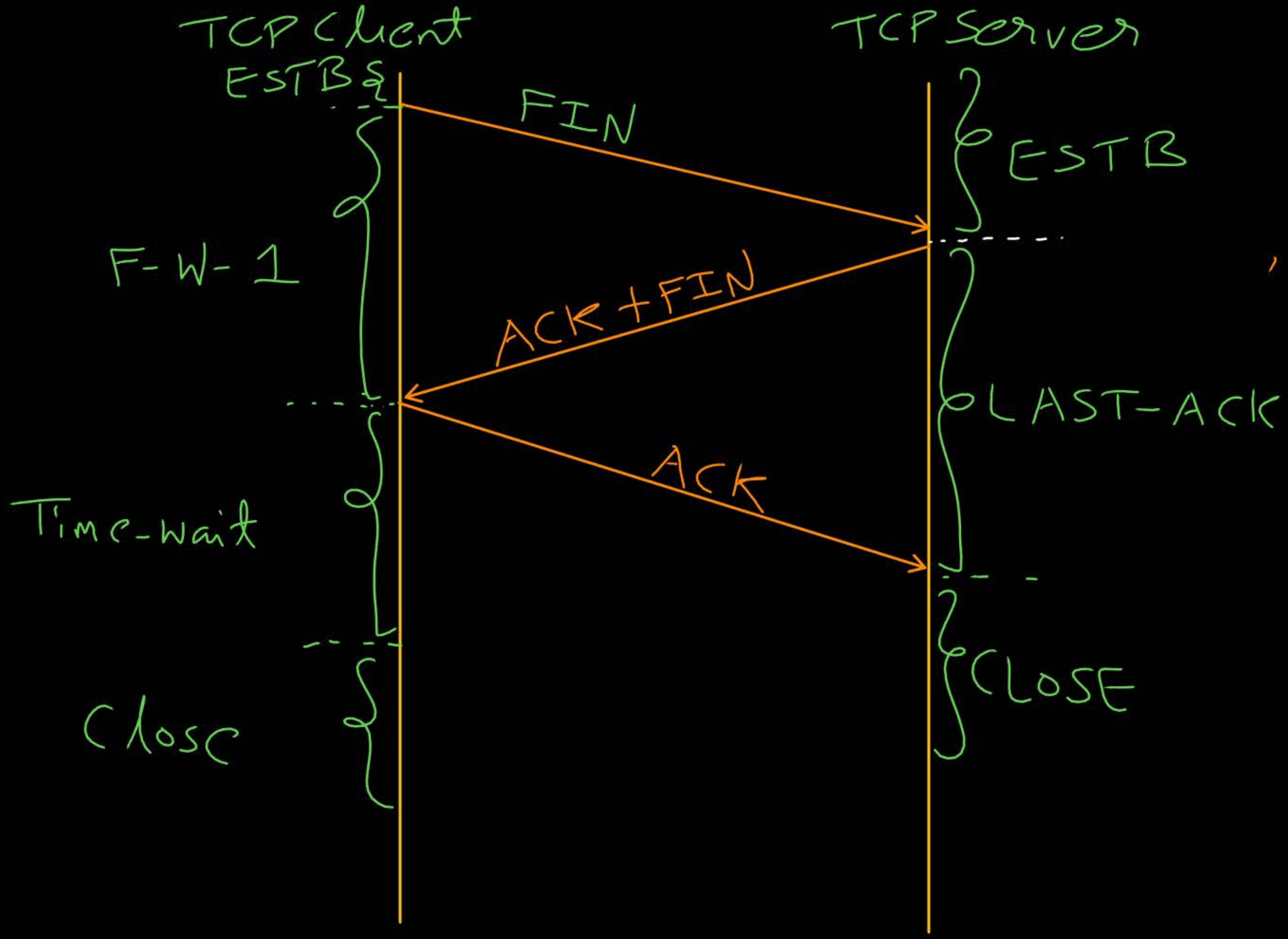
Client_ISN = 299

After sending
2000 bytes
by client.



Server_ISN = 1599

After sending
1000 bytes
by server.


TCP Client
ESTB
TCP Server
FIN
ESTB
F-W-1
ACK+FIN
LAST-ACK
ACK
Time-wait
Close
CLOSE

#Q. Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls close to terminate the connection and a FIN segment is sent to the TCP server. Server-side TCP responds by sending an ACK which is received by the client-side TCP. As per the TCP connection state diagram (RFC 793), in which state does the client side TCP connection wait for the FIN from the server-side TCP?

[GATE-2017, Set-1, 2-Mark]

(IIT-R, H.W.)

(A) LAST-ACK

(B) TIME-WAIT

(C) FIN-WAIT-1

(D) FIN-WAIT-2

2 mins Summary

Topic TCP Connection Close ✓

THANK - YOU

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